

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Flexential Portland - Hillsboro 4 Expansion, OR -- station KVVU0, America/Los_Angeles
Committed pair OSM 918873142 -> 918871787
Flexential Portland - Hillsboro 3 / QTS Data Center - Hillsboro 2
Facade gap 99.4 m between the two halls
Weather record 42,062 real hourly records from KVVU0
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3277 C at 235 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-02-28 (clear_cool)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 4 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 8 more -- but declared 0 unsafe hours against the agent's 0. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 4 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	7.223	18.0	7.220	4.914
01	mechanical	yes	-	7.232	18.0	7.780	4.364
02	mechanical	yes	-	7.232	18.0	7.780	3.633
03	mechanical	no	refusal	7.234	18.0	7.220	3.121
04	mechanical	yes	-	7.774	18.0	7.220	3.207
05	mechanical	yes	-	7.792	18.0	7.220	2.746
06	mechanical	yes	-	7.792	18.0	7.220	2.014
07	mechanical	no	refusal	9.606	18.0	7.220	2.046
08	FREE	yes	-	11.122	18.0	8.890	2.378
09	FREE	yes	-	12.775	18.0	10.000	3.247
10	FREE	yes	-	15.008	18.0	10.560	4.067

11	FREE	yes	-	17.225	18.0	11.110	4.994
12	mechanical	no	refusal	18.900	18.0	11.670	5.767
13	mechanical	no	refusal	18.899	18.0	12.942	5.500
14	mechanical	no	refusal	20.021	18.0	13.890	5.275
15	mechanical	no	refusal	20.557	18.0	14.073	5.444
16	mechanical	no	dry-bulb	19.462	18.0	12.780	5.043
17	mechanical	yes	switch budget	17.792	18.0	11.110	4.403
18	mechanical	yes	switch budget	15.572	18.0	9.440	3.487
19	mechanical	yes	switch budget	13.912	18.0	10.560	2.596
20	mechanical	no	refusal	13.594	18.0	8.330	2.649
21	mechanical	no	refusal	13.338	18.0	8.330	3.794
22	mechanical	yes	-	12.225	18.0	7.780	4.402
23	mechanical	yes	-	10.565	18.0	6.670	4.607

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.223 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.232 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.232 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.774 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.792 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more.

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06:00 MECHANICAL

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07:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.122 C, which is 6.878 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.378 C, and the margin is measured, not chosen: 2.367 C of group-conditional forecast error for this hour of day, plus 0.0116 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.775 C, which is 5.225 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.247 C, and the margin is measured, not chosen: 3.242 C of group-conditional forecast error for this hour of day, plus 0.0051 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.007 C, which is 2.993 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.067 C, and the margin is measured, not chosen: 4.059 C of group-conditional forecast error for this hour of day, plus 0.0075 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.225 C, which is 0.775 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.993 C, and the margin is measured, not chosen: 4.988 C of group-conditional forecast error for this hour of day, plus 0.0051 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a

figure it cannot stand behind.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.462 C against a 18.0 C limit, so it fails by 1.462 C. It would take a limit 1.462 C higher, or a bound 1.462 C tighter, to change this hour.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.792 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.572 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.912 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

21:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

22:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.225 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.565 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison:

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